2.38 Integration: Partial Fractions

Worked Solutions 1–14

1 a)

$\dfrac{8x+23}{x+2} \equiv A + \dfrac{B}{x+2} \Rightarrow 8x+23 \equiv A(x+2)+B$. Let $x=-2$: $7=B$. Let $x=0$: $23=2A+7 \Rightarrow A=8$. $\therefore \dfrac{8x+23}{x+2} \equiv 8 + \dfrac{7}{x+2}$.

1 b)

$$\int \frac{8x+23}{x+2}\,dx = \int \left(8 + \frac{7}{x+2}\right) dx = 8x + 7\ln|x+2| + c.$$

2 a)

$\dfrac{2x^2-2x-1}{x+1} \equiv Ax + B + \dfrac{C}{x+1}$, $2x^2-2x-1 \equiv Ax(x+1)+B(x+1)+C$. Let $x=-1$: $3=C$. Let $x=0$: $-1=B+3 \Rightarrow B=-4$. Let $x=1$: $-1=2A+2(-4)+3 \Rightarrow A=2$. $\therefore \dfrac{2x^2-2x-1}{x+1} \equiv 2x-4+\dfrac{3}{x+1}$.

2 b)

$$\int \frac{2x^2-2x-1}{x+1}\,dx = \int \left(2x-4+\frac{3}{x+1}\right) dx = x^2-4x+3\ln|x+1|+c.$$

3 a)

$\dfrac{4}{(x-3)(x+1)} \equiv \dfrac{A}{x-3} + \dfrac{B}{x+1}$, so $4 \equiv A(x+1)+B(x-3)$. Let $x=3$: $4=4A \Rightarrow A=1$. Let $x=-1$: $4=-4B \Rightarrow B=-1$. $\therefore \dfrac{4}{(x-3)(x+1)} \equiv \dfrac{1}{x-3} - \dfrac{1}{x+1}$.

3 b)

$$\int \frac{4}{(x-3)(x+1)}dx = \int \left(\frac{1}{x-3} - \frac{1}{x+1}\right) dx = \ln|x-3| - \ln|x+1| + c = \ln\left|\frac{x-3}{x+1}\right| + c.$$

4 a)

$\dfrac{10}{25-x^2} = \dfrac{10}{(5-x)(5+x)} \equiv \dfrac{A}{5-x} + \dfrac{B}{5+x}$, $10 \equiv A(5+x)+B(5-x)$. Let $x=5$: $10=10A \Rightarrow A=1$. Let $x=-5$: $10=10B \Rightarrow B=1$.

$$\begin{aligned}\int \frac{10}{25-x^2}dx &= \int \left(\frac{1}{5-x} + \frac{1}{5+x}\right) dx \\ &= -\ln|5-x| + \ln|5+x| + c = \ln\left|\frac{5+x}{5-x}\right| + c.\end{aligned}$$

4 b)

$$\int_{-1}^{1} \frac{10}{25-x^2}dx = \left[\ln\left|\frac{5+x}{5-x}\right|\right]_{-1}^{1} = \ln\frac{6}{4} - \ln\frac{4}{6}$$
$$= \ln\left(\frac{6}{4}\times\frac{6}{4}\right) = \ln\frac{36}{16} = \ln\frac{9}{4}.$$

5 a)

$\frac{2x-1}{(x+5)(x-6)} \equiv \frac{A}{x+5}+\frac{B}{x-6}$, $2x-1 \equiv A(x-6)+B(x+5)$. Let $x=-5$: $-11=-11A \Rightarrow A=1$. Let $x=6$: $11=11B \Rightarrow B=1$.

$$\int \frac{2x-1}{(x+5)(x-6)}dx = \int\left(\frac{1}{x+5}+\frac{1}{x-6}\right)dx = \ln|x+5|+\ln|x-6|+c.$$

5 b)

$$\int_7^k \frac{2x-1}{(x+5)(x-6)}dx = \ln 5$$
$$[\ln|x+5|+\ln|x-6|]_7^k = \ln 5$$
$$\ln|k+5|+\ln|k-6|-(\ln|7+5|+\ln|7-6|) = \ln 5$$
$$\ln((k+5)(k-6))-\ln 12 = \ln 5$$
$$\ln((k+5)(k-6)) = \ln 12+\ln 5 = \ln 60$$
$$(k+5)(k-6) = 60$$
$$k^2-k-30=60 \Rightarrow k^2-k-90=0$$
$$(k-10)(k+9)=0 \Rightarrow k=10 \text{ or } k=-9.$$

As $k>0$, we must have $k=10$.

6 a)

$\frac{k-4x}{x(2-x)} \equiv \frac{A}{x}+\frac{B}{2-x}$, so $k-4x \equiv A(2-x)+Bx$. Let $x=0$: $k=2A \Rightarrow A=\frac{k}{2}$. Let $x=2$: $k-8=2B \Rightarrow B=\frac{k-8}{2}$. $\therefore \frac{k-4x}{x(2-x)} \equiv \frac{k/2}{x}+\frac{(k-8)/2}{2-x}$.

6 b)

$$\int_3^4 \frac{k-4x}{x(2-x)}dx = \int_3^4\left(\frac{k/2}{x}+\frac{(k-8)/2}{2-x}\right)dx$$
$$= \left[\frac{k}{2}\ln|x|-\frac{k-8}{2}\ln|2-x|\right]_3^4$$
$$= \left(\frac{k}{2}\ln|4|-\frac{k-8}{2}\ln|2-4|\right)-\left(\frac{k}{2}\ln|3|-\frac{k-8}{2}\ln|2-3|\right)$$
$$= \left(\frac{k}{2}\ln 4-\frac{k-8}{2}\ln 2\right)-\left(\frac{k}{2}\ln 3-\frac{k-8}{2}\ln 1\right)$$
$$= \frac{k}{2}\ln 4-\frac{k-8}{2}\ln 2-\frac{k}{2}\ln 3$$

$$= \frac{k}{2}\ln 4 - \frac{k}{2}\ln 2 + 4\ln 2 - \frac{k}{2}\ln 3$$
$$= \frac{k}{2}(\ln 4 - \ln 2 - \ln 3) + 4\ln 2 = \frac{k}{2}\ln\frac{2}{3} + 4\ln 2.$$

$$\frac{k}{2}\ln\frac{2}{3} + 4\ln 2 = \ln\frac{128}{27} \Rightarrow \frac{k}{2}\ln\frac{2}{3} = \ln\frac{128}{27} - \ln 16$$
$$\frac{k}{2}\ln\frac{2}{3} = \ln\frac{8}{27} \Rightarrow \ln\left(\frac{2}{3}\right)^{k/2} = \ln\frac{8}{27}$$
$$\left(\frac{2}{3}\right)^{k/2} = \frac{8}{27} = \left(\frac{2}{3}\right)^3 \Rightarrow \frac{k}{2} = 3 \Rightarrow k = 6.$$

7 a)

$$\frac{23x+11}{(3-x)(x+2)(3x-1)} \equiv \frac{A}{3-x} + \frac{B}{x+2} + \frac{C}{3x-1}.$$
$$23x + 11 \equiv A(x+2)(3x-1) + B(3-x)(3x-1) + C(3-x)(x+2).$$

Let $x = 3$: $80 = 40A \Rightarrow A = 2$. Let $x = -2$: $-35 = -35B \Rightarrow B = 1$. Let $x = \frac{1}{3}$: $\frac{56}{3} = \frac{56}{9}C \Rightarrow C = 3$. $\therefore \dfrac{23x+11}{(3-x)(x+2)(3x-1)} \equiv \dfrac{2}{3-x} + \dfrac{1}{x+2} + \dfrac{3}{3x-1}$.

7 b)

$$\begin{aligned}\int_1^2 \frac{23x+11}{(3-x)(x+2)(3x-1)}dx &= \int_1^2 \left(\frac{2}{3-x} + \frac{1}{x+2} + \frac{3}{3x-1}\right)dx \\ &= [-2\ln|3-x| + \ln|x+2| + \ln|3x-1|]_1^2 \\ &= (-2\ln|3-2| + \ln|2+2| + \ln|3\times 2 - 1|) \\ &\quad - (-2\ln|3-1| + \ln|1+2| + \ln|3\times 1 - 1|) \\ &= (-2\ln 1 + \ln 4 + \ln 5) - (-2\ln 2 + \ln 3 + \ln 2) \\ &= (\ln 1 + \ln 4 + \ln 5) - (\ln\tfrac{1}{4} + \ln 3 + \ln 2) \\ &= \ln 20 - \ln\frac{3}{2} = \ln\frac{40}{3}.\end{aligned}$$

8 a)

$\dfrac{4x^2+1}{x^2(2x+1)} \equiv \dfrac{A}{x} + \dfrac{B}{x^2} + \dfrac{C}{2x+1}$, $4x^2 + 1 \equiv Ax(2x+1) + B(2x+1) + Cx^2$. Let $x = -\frac{1}{2}$: $2 = \frac{1}{4}C \Rightarrow C = 8$. Let $x = 0$: $1 = B$. Let $x = 1$: $5 = 3A + 3 + 8 \Rightarrow A = -2$. $\therefore \dfrac{4x^2+1}{x^2(2x+1)} \equiv -\dfrac{2}{x} + \dfrac{1}{x^2} + \dfrac{8}{2x+1}$.

8 b)

$$\begin{aligned}\int \frac{4x^2+1}{x^2(2x+1)}dx &= \int \left(-\frac{2}{x} + \frac{1}{x^2} + \frac{8}{2x+1}\right)dx \\ &= -2\ln|x| - \frac{1}{x} + 4\ln|2x+1| + c \\ &= -\frac{1}{x} + \ln(2x+1)^4 - \ln x^2 + c \\ &= -\frac{1}{x} + \ln\left(\frac{(2x+1)^4}{x^2}\right) + c.\end{aligned}$$

9 a)

$\dfrac{4x-2}{(x-2)^2} \equiv \dfrac{A}{x-2}+\dfrac{B}{(x-2)^2}$, $4x-2 \equiv A(x-2)+B$. Let $x=2$: $6=B$. Let $x=0$: $-2=-2A+6 \Rightarrow A=4$. $\therefore \dfrac{4x-2}{(x-2)^2} \equiv \dfrac{4}{x-2}+\dfrac{6}{(x-2)^2}$.

9 b)

$$\begin{aligned}\int_3^4 \frac{4x-2}{(x-2)^2}dx &= \int_3^4 \left(\frac{4}{x-2}+\frac{6}{(x-2)^2}\right)dx = \left[4\ln|x-2|-\frac{6}{x-2}\right]_3^4 \\ &= \left(4\ln|4-2|-\frac{6}{4-2}\right)-\left(4\ln|3-2|-\frac{6}{3-2}\right) \\ &= (4\ln 2-3)-(4\ln 1-6) = 4\ln 2-3+6 = 3+4\ln 2 = 3+\ln 16.\end{aligned}$$

10 a)

$$\frac{10x^2+6}{(x+1)^2(3x-1)} \equiv \frac{A}{x+1}+\frac{B}{(x+1)^2}+\frac{C}{3x-1}.$$

$$10x^2+6 \equiv A(x+1)(3x-1)+B(3x-1)+C(x+1)^2.$$

Let $x=-1$: $16=-4B \Rightarrow B=-4$. Let $x=\frac{1}{3}$: $\frac{64}{9}=\frac{16}{9}C \Rightarrow C=4$. Let $x=0$: $6=-A+4+4 \Rightarrow A=2$. $\therefore \dfrac{10x^2+6}{(x+1)^2(3x-1)} \equiv \dfrac{2}{x+1}-\dfrac{4}{(x+1)^2}+\dfrac{4}{3x-1}$.

10 b)

$$\begin{aligned}\int_1^{5/3} \frac{10x^2+6}{(x+1)^2(3x-1)}dx &= \int_1^{5/3}\left(\frac{2}{x+1}-\frac{4}{(x+1)^2}+\frac{4}{3x-1}\right)dx \\ &= \left[2\ln|x+1|+\frac{4}{x+1}+\frac{4}{3}\ln|3x-1|\right]_1^{5/3} \\ &= \left(2\ln|\tfrac{5}{3}+1|+\frac{4}{5/3+1}+\frac{4}{3}\ln|3\times\tfrac{5}{3}-1|\right) \\ &\quad -\left(2\ln|1+1|+\frac{4}{1+1}+\frac{4}{3}\ln|3\times 1-1|\right) \\ &= \left(2\ln\frac{8}{3}+\frac{3}{2}+\frac{4}{3}\ln 4\right)-\left(2\ln 2+2+\frac{4}{3}\ln 2\right) \\ &= \left(2(\ln 8-\ln 3)+\frac{3}{2}+\frac{4}{3}\ln 4\right)-\left(2\ln 2+2+\frac{4}{3}\ln 2\right) \\ &= \left(2\ln 8-2\ln 3+\frac{3}{2}+\frac{4}{3}\ln 4\right)-\left(2\ln 2+2+\frac{4}{3}\ln 2\right) \\ &= \left(6\ln 2-2\ln 3+\frac{3}{2}+\frac{8}{3}\ln 2\right)-\left(2\ln 2+2+\frac{4}{3}\ln 2\right) \\ &= \left(\frac{26}{3}\ln 2-2\ln 3+\frac{3}{2}\right)-\left(\frac{10}{3}\ln 2+2\right) \\ &= -\frac{1}{2}+\frac{16}{3}\ln 2-\ln 9.\end{aligned}$$

11)

$\frac{x^2+5x+6}{x^2(2-x)} = 0 \Rightarrow x^2+5x+6 = 0 \Rightarrow (x+2)(x+3) = 0 \Rightarrow x = -2$ or $x = -3$. $\frac{x^2+5x+6}{x^2(2-x)} \equiv \frac{A}{x} + \frac{B}{x^2} + \frac{C}{2-x}$, $x^2+5x+6 \equiv Ax(2-x) + B(2-x) + Cx^2$. Let $x = 0$: $6 = 2B \Rightarrow B = 3$. Let $x = 2$: $20 = 4C \Rightarrow C = 5$. Let $x = 1$: $12 = A + 3 + 5 \Rightarrow A = 4$. $\therefore \frac{x^2+5x+6}{x^2(2-x)} \equiv \frac{4}{x} + \frac{3}{x^2} + \frac{5}{2-x}$.

$$\begin{aligned}
\int_{-3}^{-2} \frac{x^2+5x+6}{x^2(2-x)} dx &= \int_{-3}^{-2} \left(\frac{4}{x} + \frac{3}{x^2} + \frac{5}{2-x}\right) dx \\
&= \left[4\ln|x| - \frac{3}{x} - 5\ln|2-x|\right]_{-3}^{-2} \\
&= \left(4\ln|-2| - \frac{3}{-2} - 5\ln|2-(-2)|\right) \\
&\quad - \left(4\ln|-3| - \frac{3}{-3} - 5\ln|2-(-3)|\right) \\
&= (4\ln 2 + \tfrac{3}{2} - 5\ln 4) - (4\ln 3 + 1 - 5\ln 5) \\
&= 4\ln 2 - 5\ln 4 - 4\ln 3 + 5\ln 5 + \frac{1}{2} \\
&= \ln 16 - \ln 1024 - \ln 81 + \ln 3125 + \frac{1}{2} \\
&= \ln\left(\frac{16\times 3125}{1024\times 81}\right) + \frac{1}{2} = \ln\frac{3125}{5184} + \frac{1}{2}.
\end{aligned}$$

$\therefore$ Area $= -\left(\ln\frac{3125}{5184} + \frac{1}{2}\right) = \ln\frac{5184}{3125} - \frac{1}{2}$.

12)

$$\frac{360x^2+157x+16}{(3x+1)(2-5x)(8x+3)} \equiv \frac{A}{3x+1} + \frac{B}{2-5x} + \frac{C}{8x+3}.$$

$$360x^2+157x+16 \equiv A(2-5x)(8x+3) + B(3x+1)(8x+3) + C(3x+1)(2-5x).$$

Let $x = -\frac{1}{3}$: $\frac{11}{3} = \frac{11}{9}A \Rightarrow A = 3$. Let $x = \frac{2}{5}$: $\frac{682}{5} = \frac{341}{25}B \Rightarrow B = 10$. Let $x = -\frac{3}{8}$: $\frac{31}{4} = -\frac{31}{64}C \Rightarrow C = -16$. $\therefore \frac{360x^2+157x+16}{(3x+1)(2-5x)(8x+3)} \equiv \frac{3}{3x+1} + \frac{10}{2-5x} - \frac{16}{8x+3}$.

$$\begin{aligned}
\lim_{\delta x\to 0}\sum_{x=0}^{1/8} \frac{360x^2+157x+16}{(3x+1)(2-5x)(8x+3)}\delta x &= \int_0^{1/8} \frac{360x^2+157x+16}{(3x+1)(2-5x)(8x+3)} dx \\
&= \int_0^{1/8} \left(\frac{3}{3x+1} + \frac{10}{2-5x} - \frac{16}{8x+3}\right) dx \\
&= \left[\ln|3x+1| - 2\ln|2-5x| - 2\ln|8x+3|\right]_0^{1/8} \\
&= \left(\ln|3\times\tfrac{1}{8}+1| - 2\ln|2-5\times\tfrac{1}{8}| - 2\ln|8\times\tfrac{1}{8}+3|\right) \\
&\quad - (\ln|3\times 0+1| - 2\ln|2-5\times 0| - 2\ln|8\times 0+3|) \\
&= (\ln\tfrac{11}{8} - 2\ln\tfrac{11}{8} - 2\ln 4) - (\ln 1 - 2\ln 2 - 2\ln 3) \\
&= (-\ln\tfrac{11}{8} - 2\ln 4) - (-2\ln 2 - 2\ln 3) \\
&= (\ln\tfrac{8}{11} - \ln 16) - (\ln\tfrac{1}{4} - \ln 9) \\
&= \ln\frac{1}{22} - \ln\frac{1}{36} = \ln\frac{18}{11}.
\end{aligned}$$

13)

$u = x^{1/2} - 2 \Rightarrow \dfrac{du}{dx} = \frac{1}{2}x^{-1/2} \Rightarrow dx = 2x^{1/2}du$. $u = x^{1/2} - 2 \Rightarrow x^{1/2} = u + 2 \Rightarrow dx = 2(u+2)du$.

$$\begin{aligned}\int \frac{9}{2(\sqrt{x}-2)(\sqrt{x}+1)(\sqrt{x}+4)}dx &= \int \frac{9}{2u((u+2)+1)((u+2)+4)} \times 2(u+2)du \\ &= \int \frac{9(u+2)}{u(u+3)(u+6)}du.\end{aligned}$$

$\dfrac{9(u+2)}{u(u+3)(u+6)} \equiv \dfrac{A}{u} + \dfrac{B}{u+3} + \dfrac{C}{u+6}$, $9(u+2) \equiv A(u+3)(u+6) + Bu(u+6) + Cu(u+3)$. Let $u = 0$: $18 = 18A \Rightarrow A = 1$. Let $u = -3$: $-9 = -9B \Rightarrow B = 1$. Let $u = -6$: $-36 = 18C \Rightarrow C = -2$.
$\therefore \dfrac{9(u+2)}{u(u+3)(u+6)} \equiv \dfrac{1}{u} + \dfrac{1}{u+3} - \dfrac{2}{u+6}$.

$$\begin{aligned}\int \frac{9(u+2)}{u(u+3)(u+6)}du &= \int \left(\frac{1}{u} + \frac{1}{u+3} - \frac{2}{u+6}\right)du \\ &= \ln|u| + \ln|u+3| - 2\ln|u+6| + c \\ &= \ln|\sqrt{x}-2| + \ln|\sqrt{x}-2+3| - 2\ln|\sqrt{x}-2+6| + c \\ &= \ln|\sqrt{x}-2| + \ln|\sqrt{x}+1| - 2\ln|\sqrt{x}+4| + c.\end{aligned}$$

14)

$u = (x-2)^{1/2} \Rightarrow \dfrac{du}{dx} = \frac{1}{2}(x-2)^{-1/2} \Rightarrow dx = 2(x-2)^{1/2}du = 2u\,du$. $u = (x-2)^{1/2} \Rightarrow u^2 = x - 2$.
Change limits: $x = 6 \Rightarrow u = (6-2)^{1/2} = 2$; $x = 3 \Rightarrow u = (3-2)^{1/2} = 1$.

$$\int_3^6 \frac{6}{(3+2\sqrt{x-2})(x-2)}dx = \int_1^1 \frac{6}{(3+2u)u^2} \times 2u\,du = \int_1^1 \frac{12}{(3+2u)u}du.$$

$\dfrac{12}{(3+2u)u} \equiv \dfrac{A}{3+2u} + \dfrac{B}{u}$, so $12 \equiv Au + B(3+2u)$. Let $u = -\frac{3}{2}$: $12 = -\frac{3}{2}A \Rightarrow A = -8$. Let $u = 0$: $12 = 3B \Rightarrow B = 4$.

$$\begin{aligned}\int_1^1 \frac{12}{(3+2u)u}du &= \int_1^1 \left(-\frac{8}{3+2u} + \frac{4}{u}\right)du \\ &= \left[-4\ln|3+2u| + 4\ln|u|\right]_1^2 \\ &= (-4\ln|3+2\times 2| + 4\ln|2|) - (-4\ln|3+2\times 1| + 4\ln|1|) \\ &= (-4\ln 7 + 4\ln 2) - (-4\ln 5 + 4\ln 1) \\ &= 4\ln 2 - 4\ln 7 + 4\ln 5 = 4(\ln 2 - \ln 7 + \ln 5) \\ &= 4\ln\left(\frac{2\times 5}{7}\right) = 4\ln\frac{10}{7}.\end{aligned}$$